Task 5 — Healthcare Predictive Analytics

Virtual Works Lab | B.Pharmacy Internship Project

Project Objective

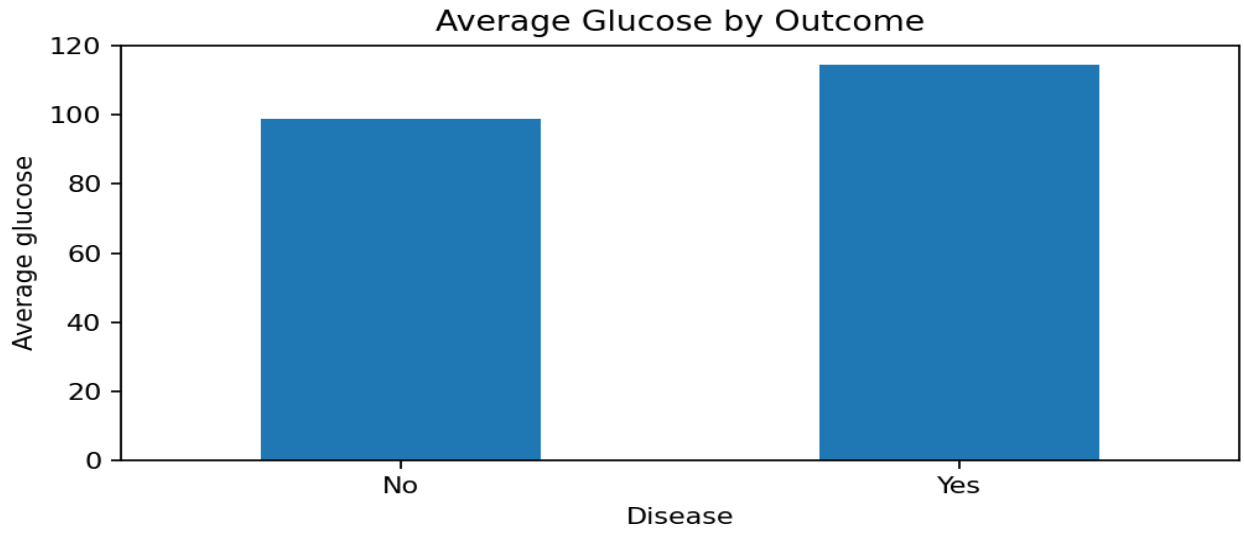
Develop a basic predictive analytics project using historical healthcare data. This project demonstrates disease prediction using a simple logistic-regression model.

Dataset

Synthetic educational dataset of 500 simulated healthcare records containing age, BMI, glucose, systolic blood pressure, family history, activity hours per week, and a binary disease outcome.

1. Explore the Historical Data

Age	BMI	Glucose	Systolic_BP	Family_History	Activity_Hours_Week	Disease
500.0	500.0	500.0	500.0	500.0	500.0	500.0
49.29	25.89	105.2	128.12	0.32	3.57	0.4
18.37	3.94	21.3	15.26	0.47	1.43	0.49
18.0	13.9	65.0	90.0	0.0	0.0	0.0
33.0	23.38	90.12	118.4	0.0	2.6	0.0
50.0	25.9	105.9	128.4	0.0	3.6	0.0
66.0	28.4	119.72	138.93	1.0	4.5	1.0
80.0	37.9	166.7	175.7	1.0	7.0	1.0



2. Predictive Model

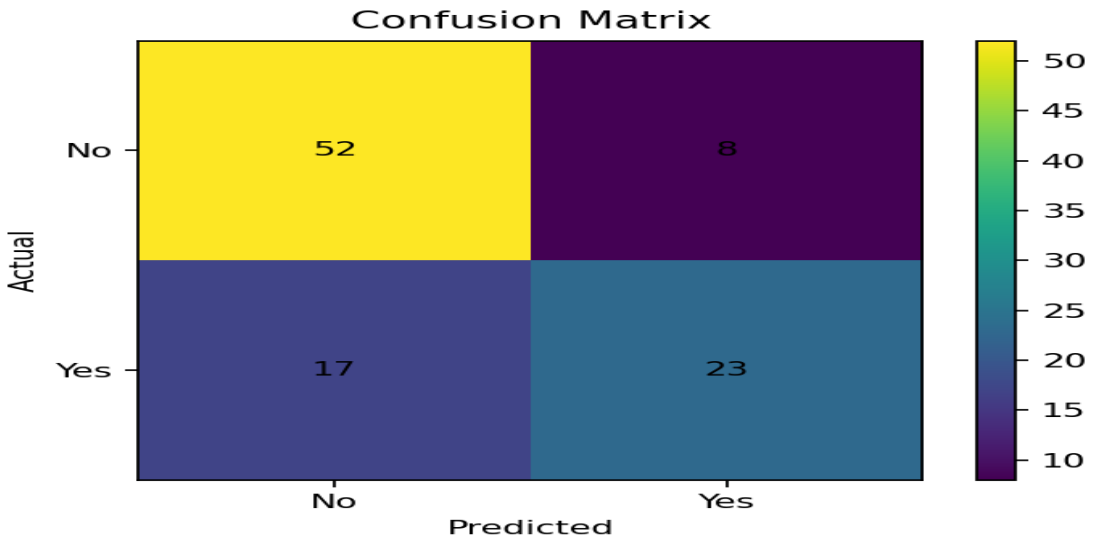
A logistic-regression classifier was trained after standardizing the input features. The data were split into 80% training and 20% testing sets using stratification.

• Test-set accuracy: 0.750 (75.0%).

• Test records: 100; training records: 400.

3. Model Evaluation

Metric	Value
Accuracy	0.75
Precision (No)	0.754
Recall (No)	0.867
F1 (No)	0.806
Precision (Yes)	0.742
Recall (Yes)	0.575
F1 (Yes)	0.648



Model Coefficients

Feature	Standardized_Coefficient
Activity_Hours_Week	-0.402
Family_History	0.058
BMI	0.292
Systolic_BP	0.463
Age	0.703
Glucose	0.979

• Positive coefficients increase the model's predicted probability in this simulated dataset; negative coefficients decrease it.

• Accuracy should be interpreted alongside precision, recall, and F1-score.

• The model is for internship demonstration only and must not be used for real patient diagnosis or treatment.

Conclusion

The project demonstrates the complete predictive-analytics workflow: data preparation, preprocessing, model training, prediction, and evaluation. Real healthcare deployment would require validated datasets, external validation, fairness checks, privacy controls, clinical oversight, and regulatory review.